

OSPF

At the end of this episode, I will be able to:

1. Open Shortest Path First (OSPF)

Learner Objective: *Learn about the OSPF dynamic routing protocol*

Description: In this episode, you will learn about the dynamic, link-state routing protocol named OSPF. Here you will learn the concepts behind this protocol and you will even see its configuration on Cisco routers.

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- OSPF is an Interior Gateway Protocol (IGP) used to route WITHIN an Autonomous System (AS)
 - OSPF is a departure from earlier protocols like RIP that are based on the Bellman-Ford vector based algorithm (routing by rumor)
 - OSPF began as an open standard and its development to this day continues that status
 - OSPF is based on the concept of a link-state protocol; links can be thought of as the interfaces of network devices; routers share these link descriptions and state information with their peers to form a "map" of the network; this "map" of the network is termed the link state database
 - Compared to RIP (specifically RIPv2) - OSPF provides many advantages including:
 - No hop count limit
 - Support for Variable Length Subnet Masking (VLSM)
 - Updates send using multicast technology (not broadcast)
 - Superior convergence times
 - Superior load balancing
 - Permits the creation of areas - these areas allow for the optimization of information sharing and more efficient operations
 - Supports the use of authentication (various strengths)
 - Easily supports the injection and tagging of external routes from other protocols (and also static routes)
 - OSPF uses the Shortest Path First algorithm in its operation - this is very efficient and allows routers to calculate the shortest paths (based on cost (bandwidth)) to network destination
 - The metric used by OSPF is cost - cost is based on link bandwidth - the shortest path will be the lowest cost and should be the highest bandwidth path
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Additional Resources:

Understand Open Shortest Path First (OSPF) - Design Guide

<https://www.cisco.com/c/en/us/support/docs/ip/open-shortest-path-first-ospf/7039-1.html>